

Can lizards count? Testing numerosity in an Australian skink using spontaneous quantity discrimination

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Quantity discrimination is a cognitive trait that allows animals to process numerical information. Although well studied in mammals, birds and fishes, little is known about quantity discrimination in nonavian reptiles. Quantity discrimination is typically measured through spontaneous binary choice experiments or by training animals to discriminate quantities through conditioning, but rarely are both methods used in tandem. We examined spontaneous quantity discrimination with and without positive reinforcement in the eastern water skink, *Eulamprus quoyii*. We offered a series of binary choices: one versus four, two versus four, three versus four, two versus eight, four versus eight and six versus eight same-sized crickets for number discrimination and 0.3 versus 1.2 cm, 0.6 versus 2.4 cm, 0.6 versus 1.2 cm, 1.2 versus 2.4 cm, 0.9 versus 1.2 cm and 1.8 versus 2.4 cm for size discrimination tests. We quantified number discrimination performance using crickets of the same size that varied in number and size discrimination performance using two crickets that varied in size. These tests helped determine which core system the skinks use and assessed the effect of positive reinforcement. In the spontaneous experiment, we found that skinks could only discriminate between one versus four crickets and 0.3 versus 1.2 cm. However, when rewarded, skinks successfully discriminated one versus four crickets in number and between all ratios in size, except for the highest ratio of 1.8 versus 2.4 cm. These results indicate a possible difference in discrimination method when faced with number or size and showed that training can improve an animal's discrimination capacity.

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Quantity discrimination (QD) enables animals to process information about quantity when faced with discrete and continuous numerical information, such as the number or relative size of objects (Agrillo & Bisazza, 2018; Nieder, 2020). According to optimal foraging theory, when animals are faced with a choice between two or more options, they should select the one that offers the greatest benefit (Kelly, 2016; Pyke, 1984). Therefore, animals must be able to distinguish between different amounts or sizes of a resource to make the most optimal decisions for their survival and fitness. Fishes rely on QD to compare the relative number or size of different shoals to decide which group to join (Agrillo et al., 2007; Gómez-Laplaza & Gerlai, 2011). Birds use QD to track the number of eggs in their nest or recall how much food they have stored within caches (Garland et al., 2012; Hunt et al.,

2008; Lyon, 2003). Mammals, such as African lions, *Panthera leo*, use QD in a social context to decide how best to respond to intruders (McComb et al., 1994). In humans, QD forms the basis of our understanding of numbers and mathematics (Feigenson et al., 2004; Hyde, 2011). Thus, being able to distinguish between quantities is predicted to have fitness benefits across multiple contexts.

Two core systems within the brain allow for the processing of quantity information. These systems are known as the object file system (OFS) and the approximate number system (ANS; Hyde, 2011). The OFS allows for the precise tracking of individual items within sets but is commonly limited to small numbers (typically up to three or four). The ANS encodes the approximate magnitudes of items within sets and does not appear to have an upper limit; instead, it obeys Weber's law, where accuracy decreases as the distance between numbers decreases (Feigenson et al., 2004; Hyde, 2011). Based on the differences in the limits of the two systems, it would be logical to predict that the brain decides which

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system to use based on the numerical range to be distinguished. However, studies have shown that the ANS can be used for small quantities (Cordes et al., 2001; Hyde, 2011; Rugani et al., 2013), and others have found that some animals can use the OFS beyond its typical limit of four items (Garland et al., 2012; Kelly, 2016). The use of either system may not be dictated by numerosity but rather by other factors, such as the availability of continuous information or the context of the task (Balestrieri et al., 2019; Rugani, 2018; Stancher et al., 2015). Although plausible, these factors have not been widely tested in the context of understanding the conditions that will trigger the use of one system over the other.

Two main methods are used to study QD: spontaneous QD and trained QD. Spontaneous QD presents naïve animals with binary choices of more versus less or larger versus smaller (Agrillo & Bisazza, 2014). No prior learning is conducted before presenting animals with these choices. Trained QD also presents animals with binary choices but is preceded by training the animals to an association or rule, typically to consistently select the smaller or larger option through two-choice associative learning tasks (Agrillo et al., 2012; Piffer, Miletto Petrazzini, & Agrillo, 2013; Vila Pouca et al., 2019). Spontaneous QD can be useful for studying the ecological relevance of QD but is commonly limited by the animal's motivation (Agrillo & Bisazza, 2014). Alternatively, trained QD provides an avenue for testing QD within a more controlled and standardized setting and avoids inconsistencies in motivation by training animals to obtain a reward (Agrillo et al., 2012; Piffer et al., 2013; Vila Pouca et al., 2019). Although limited in its ability to evaluate an animal's cognitive performance in a natural setting, this method has been shown to increase an animal's QD capacity, which allows researchers to understand the absolute cognitive upper limit of an animal's QD ability (Agrillo & Bisazza, 2014; Bisazza et al., 2014; De Long et al., 2017).

Many studies have investigated QD in mammals (Cantlon et al., 2009; Neider, 2020), birds (Agrillo & Bisazza, 2014; Rugani, 2018), fishes (Agrillo & Bisazza, 2014, 2018) and even invertebrates (Gatto et al., 2022). However, QD is poorly understood in nonavian reptiles, with only a handful of studies to date. Within this group, the Italian wall lizard, *Podarcis sicula* (Miletto Petrazzini et al., 2017, 2018), Hermann's tortoise, *Testudo hermanni* (Gazzola et al., 2018), Cyren's rock lizard, *Iberolacerta cyreni* (Recio et al., 2021), Chinese stripe-necked turtle, *Mauremys sinensis* (Lin et al., 2021), shingleback skink, *Tiliqua rugosa* (Szabo et al., 2023) and red-eared slider, *Trachemys scripta elegans* (Sun et al., 2023) appear to use the ANS. In contrast, the gidgee skink, *Egernia stokesii*, is the only studied nonavian reptile that used the OFS during experiments (Szabo et al., 2021). Whether all nonavian reptile species possess both core systems but only one is revealed within a given experimental condition or if they exclusively possess a single core system remains largely unknown.

Here, we investigated QD performance in the eastern water skink, *Eulamprus quoyii*, using tests of spontaneous QD with and without positive reinforcement. Water skinks are sit-and-wait foragers that rely on vision for detecting prey, and previous research has shown that they respond to cognitive tasks using visual cues (Noble et al., 2014; Qi et al., 2018; Szabo et al., 2019, 2020; Veron, 1969). These features make them ideal candidates for participating in QD experiments that rely on visual discrimination. An ability to count or discriminate between differently sized prey would be advantageous in making decisions regarding finding and catching food. Our methods follow those used in two previous studies on the gidgee skink, *E. stokesii*, and shingleback skink, *T. rugosa*, to facilitate better comparisons between species (Szabo et al., 2021, 2023). We used spontaneous QD to reflect decision-making in the wild, followed by positive reinforcement training to understand the absolute upper limit of the skink's QD abilities.

By presenting the water skinks with binary choice tasks dealing with different amounts and sizes, we aimed to understand how they use the two core systems and how these may differ based on experimental protocol.

We used ratios of 0.25, 0.50 and 0.75 to identify the core system that the skinks use. If the water skinks choose the larger (or smaller) option with decreasing accuracy as the ratio increases, this trend would indicate that they use the ANS. However, if their accuracy is not affected by ratio and is limited by an absolute number, this result would suggest that they use the OFS. We used the size discrimination test to understand the effect of discrete (number) compared to continuous (size) discrimination and to investigate how differences in the available information may affect the water skink's discrimination abilities. Better accuracy when discriminating one type of information compared to the other could suggest a greater importance in the availability of that information.

METHODS

Study Species

The eastern water skink is a nongroup-living, diurnal species (Law & Bradley, 1990; Noble et al., 2014). They are often found near bodies of water, along the coast and ranges of eastern Australia, from northern Queensland to southeastern South Australia (Borges-Landáez & Shine, 2003; Pepper et al., 2018; Swan, 1990). Water skinks are insectivorous and relatively common in suburbia. Previous research has shown that they are capable of inhibitory control, social learning and spatial learning, among other cognitive tasks (Carazo et al., 2014; Noble et al., 2014; Qi et al., 2018; Szabo et al., 2019, 2020).

Animal Husbandry

For the spontaneous QD experiments, we tested nine adult male eastern water skinks on campus at Macquarie University Fauna Park. We chose to only test males to avoid any possible differences in discrimination ability based on sex, as male and female water skinks have shown differences in spatial learning ability (Carazo et al., 2014). These skinks were caught from locations across Macquarie University campus in November 2022 and housed for 4 months prior to the beginning of the study. The skinks participated in separate spatial memory experiments during the 4 months prior to the start of our experiments (Suter et al., 2022). In the positive reinforcement experiments, 12 male water skinks were caught from the same population within Macquarie University at the end of February 2024 and housed at Macquarie University Fauna Park. These skinks were given a 2-week acclimatization period to become habituated to captivity.

All individuals were identified using PIT tags. We housed the skinks individually in open 88 L plastic tubs (67 cm × 45 cm × 38.5 cm). The room temperature was maintained around 25 °C with a 12 h light cycle (from 0700 to 1900 hours). All lizards had a heat cord at one end of their enclosure, beneath the tub, to create a thermal gradient from approximately 23 °C–29 °C that allowed for thermoregulation. We monitored the skinks daily to ensure their welfare and keep them habituated to the experimenter. All enclosures contained a UVB lamp, a single refuge, a water bowl large enough for individuals to sit in and sometimes they were given enrichment items such as eucalypt branches. Skinks were kept on cat litter, which was spot cleaned weekly.

When not participating in experiments, we fed the skinks three times a week (between 0900 and 1400 hours) with dog food (Baxter's Dog Food) on Mondays, two large crickets on Wednesdays and two large crickets dusted with URS Ultimate Calcium

Powder and URS Ultimate Vitamin Powder on Fridays. During experiments, we did not feed the skinks at their usual times; instead, they were fed small crickets during trials. In the spontaneous QD experiments, skinks were fed one small cricket at the beginning of each trial. During training, skinks were fed when they approached the iPad and selected the larger option. We fed skinks the remainder of their allocated crickets at the end of a testing day if they did not receive six small crickets during experimentation. Skinks had *ad libitum* access to water.

Experimental Set-up

We created the stimuli to be displayed on an iPad (9th generation, Apple Inc., 2021) using Microsoft PowerPoint (version 16.70). We used the same image of a cricket across all stimuli, and the images were animated to move up and down to mimic the small jumping movements of crickets. These animations were created using the animation tool in PowerPoint and saved in Graphics Interchange Format (GIF). GIFs of the fake crickets were displayed using the Photos application on the iPad. All stimuli can be found at https://osf.io/ftkbu/?view_only=894e43e11eaf4aa8b993e2a73f8d15bc.

We built a wooden frame (28.5 cm × 5 cm × 26.5 cm) to ensure the iPad was consistently positioned at one end of the tub. This position was always against the wall opposite the hide that acted as the starting point (Fig. 1a). A wooden partition (16.5 cm × 1.4 cm × 21.3 cm) was attached perpendicular to the frame and divided the iPad display into two distinct sides, such that when the skink approached one half of the screen, they could not see the other half. During spontaneous QD, a small, perforated acrylic box containing live crickets was placed in the middle behind the wooden frame, centred by the wooden partition, to provide additional chemical and auditory cues and avoid concentrating the cues on one side. A small cardboard box was used to obscure the stimuli from the

skinks before the start of a trial by placing it in front of the hide entrance before inserting the iPad into the enclosure.

We attached a GoPro (Hero 7 Silver, 30 fps) to the wall above the iPad screen using an Otto Action Camera arm mount, such that the camera had a bird's eye view of the entire tub to keep track of the skink's position throughout the experiment (Fig. 1b). We observed the skinks via a live video feed using the GoPro Quik application (version 12.0.1). This set-up allowed for the observation of each skink's choice and avoided any influence the presence of a researcher may have on the skink's decisions or behaviour.

General Procedure

We conducted two experiments: a spontaneous QD test and a spontaneous QD with positive reinforcement test. A side preference test was performed before the spontaneous QD test, and a pretraining phase was conducted before the positive reinforcement experiment (details below). The spontaneous QD experiment was carried out first, although different animals were used in each test. Both tests involved number discrimination and size discrimination, which were performed simultaneously. The order of trials was randomized and intermixed for the number discrimination and size discrimination tests for all individuals. We carried out all experiments in each skink's enclosure to minimize stress due to handling.

We used 12 treatments (six for number and six for size) that covered three ratios: 0.25, 0.50 and 0.75, representing discrete (number) and continuous (size) information. These treatments were the same for both experiments. For the number discrimination test, we used different numbers of crickets: one versus four and two versus eight for ratios of 0.25, two versus four and four versus eight for ratios of 0.50 and three versus four and six versus eight for ratios of 0.75 (Table 1). All digital crickets were the same size (1.5 cm in length) but varied in number (ranging from one to



Figure 1. (a) A bird's eye view of the experimental set-up containing the hide (black box), which was the starting point approximately 20 cm from the start of the wooden partition, and the iPad screen within the wooden frame. (b) A picture of the GoPro mounted above the experimental arena to allow the researcher to observe the skinks remotely. The small acrylic box containing live crickets used during the spontaneous quantity discrimination experiment can be seen behind the wooden frame in this image. (c) A picture of the iPad within the wooden frame taken from the skink's perspective.

Table 1
Summary of the ratios used for the number and size discrimination tests in the spontaneous and trained quantity discrimination experiments

Ratio	Treatment
Number discrimination	
0.25	1 versus 4
0.25	2 versus 8
0.5	2 versus 4
0.5	4 versus 8
0.75	3 versus 4
0.75	6 versus 8
Size discrimination	
0.25	0.3 versus 1.2 cm
0.25	0.6 versus 2.4 cm
0.5	0.6 versus 1.2 cm
0.5	1.2 versus 2.4 cm
0.75	0.9 versus 1.2 cm
0.75	1.8 versus 2.4 cm

For the number discrimination test, ratios are represented as the relative number of crickets. For the size discrimination test, ratios are represented as relative sizes of crickets, ranging from 0.3 to 2.4 cm.

eight; Fig. 2a). We arranged the crickets to roughly cover the same area of the iPad screen (~9 cm across the screen on both sides with a ~2.5 cm gap between the two halves), where possible, to control for total surface area. The jumping animations for all crickets were randomly ordered but ensured that only one cricket would jump at a time on either side of the iPad screen.

For the size discrimination test, we used two digital crickets differing in size (0.3–2.4 cm) spaced approximately 8–10 cm apart, and ratios were represented by relative size (e.g. 0.5 was 0.6 cm versus 1.2 cm and 1.2 cm versus 2.4 cm; Table 1, Fig. 2b). This test would help to reveal if the skinks could perform QD using continuous information, such as total surface area. The maximum size of the cricket was limited to the largest possible cricket that the skinks were familiar with (approximately 2.4 cm in length). All digital crickets were presented on a blank white background to be consistent and

ensure that the stimuli presented were the sole basis used during discrimination. Skinks were split into two batches for ease of testing and received all 12 combinations over two testing days. We randomized the order in which the combinations were presented and included a pseudorandomized order for the side presenting the larger quantity or sized cricket to ensure that the larger option was not on the same side more than twice in a row for each skink.

Before starting an experiment, we gently ushered the skink under its hide at the opposite end of the tub, relative to the iPad, or if the skink was already in its hide, we gently moved it to the allocated starting point. Once the skink was within its hide, the cardboard box was placed at the entrance to obscure its view, and all other items in the enclosure were removed (i.e. the water bowl and enrichment items). We placed the iPad against the opposing wall, with the GoPro positioned above the set-up and the container of crickets behind the wooden frame (spontaneous QD experiment only). We left the skink under its hide, blocked by the cardboard box for a minute before the cardboard was removed. We stepped away from the tub to begin the trial.

Skinks were given up to 10 min to decide; however, if a skink had only begun to leave its hide around 10 min, it was given an additional 5 min to approach the iPad. If no movement was seen within 10 min or the skink did not choose by 15 min, the trial was ended, the experimental set-up was removed and the result of the trial was noted as 'NA' (i.e. no answer). If a choice was made, the experimental set-up was removed and the next skink was tested. We recorded each skink's choice as either selecting the 'smaller' (0) or 'larger' (1) amount/size respective to the combination presented. We defined a choice as the first stimulus the skink approached: if its snout was within 1 cm of the iPad screen, if it remained with all four legs past the wooden partition for at least 3 s or if it attempted to eat any cricket on the screen. In most cases, the skinks performed a lot of tongue-flicking behaviour and would attempt to consume the digital cricket upon approach. Before starting a subsequent trial, we mixed the substrate in the tub by hand to mix up any possible scent cues left from a previous trial. Skinks were given at least 20 min between trials to reduce stress.

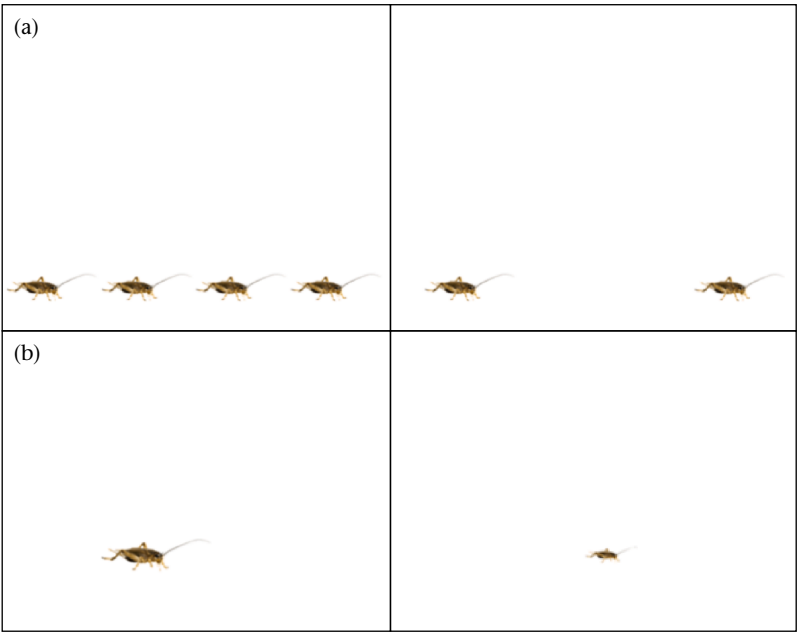


Figure 2. An example of two versus four for (a) the number discrimination test and (b) the size discrimination test. The vertical black line down the middle represents the split between the left and right sides of the screen created by the partition of the wooden frame.

Spontaneous QD: Side Preference Test

Before starting the spontaneous QD experiment, we conducted a side preference test to determine whether any skinks had significant left or right biases. We displayed two equally sized crickets (2 cm in length) on an iPad screen, spaced approximately 8 cm apart, and recorded which side the skinks chose (Fig. A1). We tested all ($N = 9$) skinks 10 times.

None of the skinks showed a significant side bias ($P > 0.05$; Table A1, Fig. A2). All skinks completed 10 trials, except for MWS003 and MWS031, which completed nine and eight trials, respectively. However, the lack of bias detected may be due to a type II error.

Spontaneous QD Experiment

At the start of a trial, a small cricket was placed at the hide's entrance to encourage the skink to exit their hide in search of more food. The skinks would anticipate the presence of more food and would more readily engage with the digital crickets, as they were habituated to being fed more than one cricket as part of their regular husbandry. The trial started when the cricket was consumed or, in the rare event that a skink did not consume the cricket, the experiment started when it first poked its entire head out of the hide.

All water skinks ($N = 9$) underwent 15 trials for each of the 12 treatments, but none completed all trials, as they did not make a choice every time. These inconsistent responses resulted in an unequal number of data points across all treatments. By the end of data collection, the skinks made choices in 1341 of a total of 1620 trials (83% of the total trials; Table A2).

Spontaneous QD with Positive Reinforcement: Pretraining Phase

Before starting the positive reinforcement experiment, we trained the skinks to approach the iPad. We displayed the same two crickets from the side preference test on the iPad screen and placed a small Petri dish filled with water containing a small cricket in the centre of the tub. The water prevented the cricket from leaving the Petri dish, and the position of the dish allowed skinks to approach the iPad with both sides of the screen visible. We repeated this procedure with the skinks ($N = 12$) 20 times. Once completed, we removed the Petri dish with the cricket and conducted another 10 repetitions in which no initial cricket was provided. We rewarded the skinks with a small cricket if they directly approached and interacted with the iPad screen, regardless of the side they chose.

Spontaneous QD with Positive Reinforcement Experiment

The experiment started when a skink first poked its entire head out of the hide. Skinks were not given a cricket at the start of the trial and were only rewarded if they chose the larger option. No reward was given if the skink chose the smaller option. Each water skink ($N = 12$) underwent 180 trials, and only five completed all trials. In total, 2086 of 2160 (97%) trials were completed.

Ethical Note

All procedures were noninvasive behavioural observations approved by the Macquarie University Animal Ethics Committee (ARA #2013/031-51). To reduce stress, lizards were handled only when absolutely necessary (e.g. taking morphological measurements) and all trials were conducted in their home enclosures. Our

study was reward-based, to which the lizards responded positively, consistent with enrichment. Lizards were returned to the wild at the end of the study. This work was performed under the New South Wales Department of Planning, Industry and Environment Licence SL101972.

Statistical Analyses

All statistical analyses were conducted in R version 4.3.3 (R Core Team, 2024). A lack of response was listed as NA (i.e. no answer) in the data set, and the associated entries were removed from the analysis. NAs were removed from the side preference and number and size discrimination data sets.

For the side preference test, we used a binomial test to determine whether skinks preferred the left or right side above chance ($P < 0.05$). For the number and size discrimination tests, we set the credible intervals to 89% for the generalized linear mixed model (GLMM) and post hoc tests as it is deemed to be more stable than the commonly used 95% (Makowski et al., 2019). We fitted a binomial Bayesian GLMM using the package 'brms' (<https://github.com/paul-buerkner/brms>, version 2.20.1, default priors, Bernoulli variable) to determine if the skinks chose the larger or smaller option above or below random chance. We included their choice (that is, picking the larger or smaller option), treatment and trial as fixed effects with a random intercept for animal identity (ID) nested in trial to account for repeated measures and autocorrelation. The highest density interval (package 'bayestestR', <https://cran.r-project.org/web/packages/bayestestR/>, version 0.13.1) was set to 89% to determine the significance of each treatment. We used the package 'emmeans' (<https://cran.r-project.org/web/packages/emmeans/>, version 1.7.4-1) to conduct post hoc least squares mean (LSM) tests and compared the estimated probability of choice accuracy between treatments. Statistical significance was assumed if the 89% credible intervals did not cross zero. Repeatability was calculated for individual choices across trials using the package 'rptR' (<https://cran.r-project.org/web/packages/rptR/index.html>, version 0.9.22) accounting for skink ID by fitting a random slope nested in trial to determine within and between individual consistency in choice preference. We assumed statistical significance for repeatability at $P < 0.05$.

RESULTS

In the spontaneous QD experiment, the skinks were only successful in discriminating between one and four for the number and size discrimination tests (LSM, marginal mean_{number} = 0.3077, CI_{lower} = 0.0566, CI_{upper} = 0.5230; LSM, marginal mean_{size} = 0.2662, CI_{lower} = 0.0433, CI_{upper} = 0.5023) and showed no significant preference when presented with any other treatments (Fig. 3a, Fig. A3). There appeared to be a weak correlation between the skink's choice during a trial and the type of discrimination test (number or size; GLMM, estimate = 0.07, CI_{lower} = 0.01, CI_{upper} = 0.18) and between the skink's choice and total trial number (GLMM, estimate = 0.04, CI_{lower} = 0.01, CI_{upper} = 0.08), suggesting that their discrimination performance was weakly affected by whether they were presented with number or size and the amount of times they had been tested.

Following positive reinforcement training, the skinks could only successfully discriminate two of the ratios presented in the number discrimination test: one versus four crickets, where they consistently chose the larger amount (LSM, marginal mean_{number} = 0.2554, CI_{lower} = 0.0647, CI_{upper} = 0.4487) and six versus eight crickets, where they consistently chose the smaller amount (LSM, marginal mean_{number} = -0.2115, CI_{lower} = -0.4020,

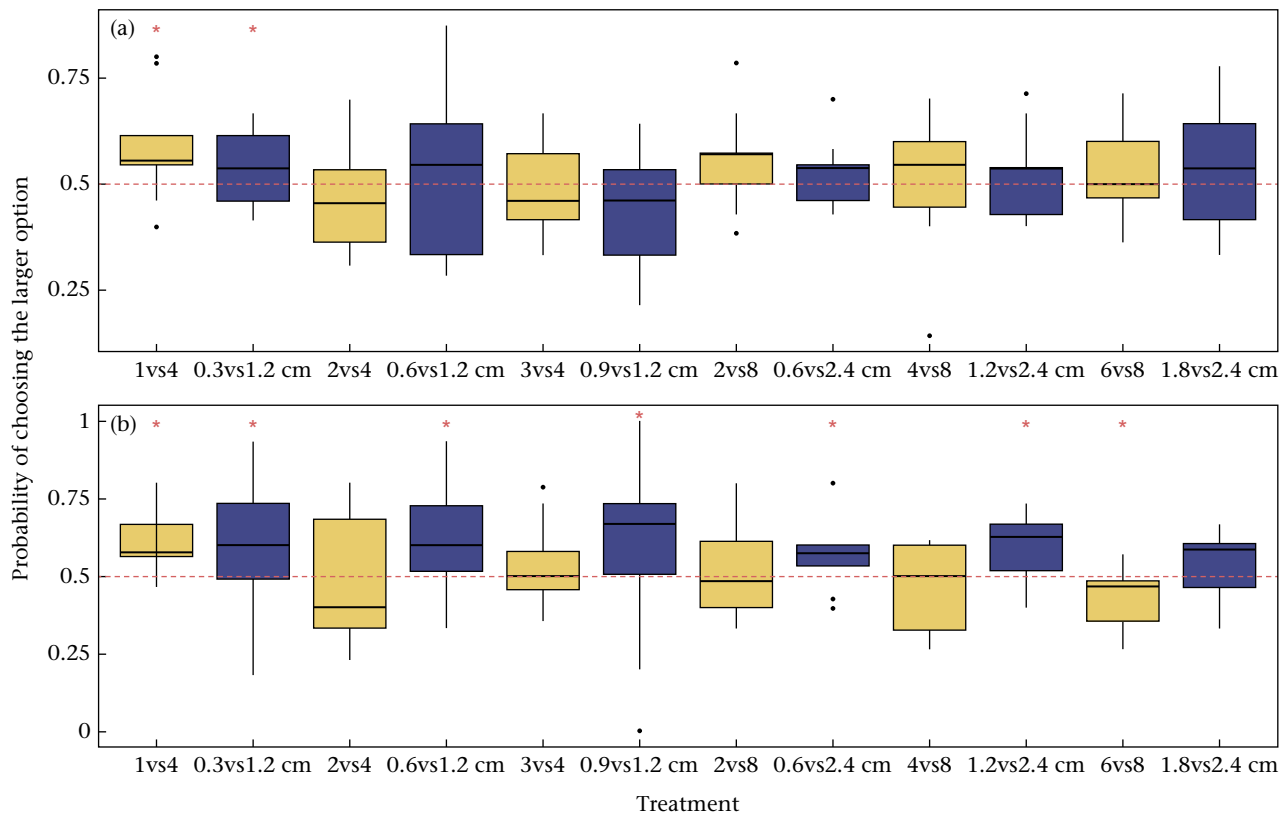


Figure 3. Box plots showing the interquartile ranges for the skinks' probability of choosing the larger option for number (yellow) and size (blue) during the (a) spontaneous and (b) trained quantity discrimination experiments. Each box represents the interquartile range showing the distribution of 50% of the data for each treatment and the vertical lines show the distribution of 25% of the data above and below the interquartile range. The horizontal black line within each box shows the median probability for each treatment and the red dotted line indicates 0.5 probability. The numbers listed for the number discrimination test reflect the number of crickets and the sizes listed for the size discrimination test reflect the sizes (cm) used in each treatment. * $P < 0.05$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

$CI_{upper} = -0.0182$). They were unable to discriminate between any other number discrimination treatments (Fig. 3b). For the six size discrimination treatments, the skinks successfully discriminated between 0.3 and 1.2 cm (LSM, marginal mean size = 0.6006, $CI_{lower} = 0.3975$, $CI_{upper} = 0.7910$), 0.6 and 2.4 cm (LSM, marginal mean size = 0.4599, $CI_{lower} = 0.2783$, $CI_{upper} = 0.6659$), 0.6 and 1.2 cm (LSM, marginal mean size = 0.3824, $CI_{lower} = 0.1845$, $CI_{upper} = 0.5660$), 1.2 and 2.4 cm (LSM, marginal mean size = 0.2707, $CI_{lower} = 0.0761$, $CI_{upper} = 0.4541$) and 0.9 and 1.2 cm (LSM, marginal mean size = 0.3316, $CI_{lower} = 0.1445$, $CI_{upper} = 0.5331$). They were unable to discriminate the highest ratio of 1.8 versus 2.4 cm (LSM, marginal mean size = 0.1325, $CI_{lower} = -0.0584$, $CI_{upper} = 0.3185$). There appeared to be a weak correlation between the skink's choice during a trial and the type of discrimination test (number or size; GLMM, estimate = 0.08, $CI_{lower} = 0.01$, $CI_{upper} = 0.18$) and between the skink's choice and total trial number (GLMM, estimate = 0.02, $CI_{lower} = 0.001$, $CI_{upper} = 0.04$).

Repeatability

Repeatability was not significant when accounting for skink ID in the spontaneous QD experiment ($P = 0.726$, $R = 0.004$, $CI_{lower} = 0$, $CI_{upper} = 0.17$) and the trained QD experiment ($P = 1$, $R = 0$, $CI_{lower} = 0$, $CI_{upper} = 0.007$), suggesting that individual choices were not repeatable across trials.

DISCUSSION

The spontaneous QD test results show that the eastern water skinks could only discriminate a ratio of 0.25 for the lowest

combinations of number and size: one versus four crickets and 0.3 versus 1.2 cm. The skinks did not consistently discriminate between any other combinations. In the spontaneous QD with positive reinforcement test, the skinks consistently discriminated between one versus four and six versus eight crickets in the number-based discriminations and could discriminate 0.3 versus 1.2 cm (ratio of 0.25), 0.6 versus 2.4 cm (ratio of 0.25), 0.6 versus 1.2 cm (ratio of 0.5), 1.2 versus 2.4 cm (ratio of 0.5) and 0.9 versus 1.2 cm (ratio of 0.75) in the size-based discriminations. However, their success in discriminating six versus eight crickets is a possible type I error, and their performance in the other discrimination tests suggests it is unlikely that they can discriminate such a challenging ratio. These results suggest that the skinks may be affected by ratio, the availability of discrete (number) and continuous (size) information within these experiments and that positive reinforcement helped to improve their discrimination performance.

The eastern water skink may rely on the ANS, which is weakly showed in the decrease in accuracy as the ratio increased in the positive reinforcement test, following Weber's law (Hyde, 2011). This pattern matches the majority of studied nonavian reptiles that also appear to use the ANS (Gazzola et al., 2018; Lin et al., 2021; Miletto Petrazzini et al., 2017, 2018; Recio et al., 2021; Sun et al., 2023; Szabo et al., 2023). Unlike the water skink, the gidgee skink, *E. stokesii*, cannot discriminate combinations when only continuous information was present (Szabo et al., 2021). Similarly, the shingle-back skink, *T. rugosa*, performs better in QD tasks when continuous and discrete information are available simultaneously, similar to human infants and eastern mosquitofish, *Gambusia holbrooki* (Agrillo et al., 2011; Cordes & Brannon, 2008; Szabo et al., 2023). Based on the results of the positive reinforcement experiment, the

skinks' inconsistencies in discriminating between two versus four in the number discrimination test could also suggest the absence of the OFS. However, whether another context may elicit the use of the OFS requires further testing, and these results cannot definitively confirm its absence.

Interestingly, the skinks did not discriminate two versus eight during the number discrimination trials in both tests despite sharing the same ratio as one versus four. Their performance may have been influenced by the higher total number of crickets (that is, 10 crickets in total for two versus eight instead of five crickets for one versus four), causing confusion between the two options. This confusion could also explain their seemingly random choices beyond one versus four during the number discrimination trials in both tests. In the case of the spontaneous QD experiment, from the skink's perspective, a lack of preference may have been adequate, especially as the total number of crickets increases. This behaviour is similar to what is seen in mice and rats, where the animals turned to guessing or showing a lack of preference when either result was deemed optimal (Kay et al., 2006). Additionally, as skinks are sit-and-wait predators, it may not be crucial to develop an ability to count beyond simply identifying an individual compared to a group, as they cannot catch multiple prey items simultaneously. It may be more advantageous to discriminate between the sizes of prey items to catch the most optimally sized individual, as seen in water skinks' better QD performance when shown ratios involving size during the positive reinforcement experiments. Similar decision-making behaviours are seen in cats, *Felis catus*, another sit-and-wait predator, who choose their prey based on a combination of size and number, as opposed to purely the amount, possibly to improve their chance of success when hunting (Bánszegi et al., 2016; Chacha et al., 2020). Future research comparing the QD abilities of closely related species with different hunting strategies would be valuable for understanding how hunting preferences affect QD ability.

An alternative explanation is that the experimental design may not have encouraged the use of either core system, particularly when faced with the number discrimination tests. Instead, the skinks may have completed the QD tasks using singular versus plural set comparisons, a possibly more efficient method of discrimination, resulting in only a successful decision for one versus four crickets, as no other treatments could be viewed in the manner of singular versus plural. Rhesus monkeys, *Macaca mulatta*, undergoing experimental conditions designed to limit the use of either core system showed the use of singular versus plural set discrimination and were only successful at discriminating combinations involving one item versus many items (that is, one versus two and one versus five; Barner et al., 2008). Further research targeting the use of singular versus plural set comparisons in the eastern water skink may be useful for understanding the experimental conditions that may trigger its use over the ANS or OFS.

Based on their foraging strategy, the water skinks would be expected to consistently pick the larger-sized cricket when performing size discrimination to maximize the fitness gain when choosing between two differently sized prey options. This prediction is supported by the positive reinforcement test results, where skinks consistently discriminated all size ratios except for the highest ratio of 1.8 versus 2.4 cm. This trend was not present in the spontaneous QD test. However, their lack of preference regarding size in the spontaneous test may reflect a lack of significant consequence for guessing as the skinks were habituated to a regular feeding schedule, resulting in poor choice accuracy beyond the easiest ratio, and may indicate their poorer performance was affected by a lack of motivation rather than a cognitive limit.

Repeated exposure to the experimental procedure would likely encourage some form of learning in the spontaneous discrimination tests. Although a cricket was given at the beginning of the

experiment, this procedure may have unintentionally taught the skinks that the reward comes from starting a trial rather than actively participating in it. Similarly, the lack of reward once a decision was made may have become a negative association, such that the absence of a reward punishes the animal. The issue of perceived punishment was discussed in a QD study on shingleback skinks, where the authors suspected that a loss of motivation over time could be attributed to a negative association with participating in experiments (Szabo et al., 2023). As such, the water skinks may have inadvertently learned that regardless of their decisions, there was no reward for accuracy and neither choice mattered during the spontaneous QD test. This learning could be the reason for the lack of individual repeatability and an observed decrease in motivation over successive testing days, suggesting that a lack of motivation rather than cognitive limitations influenced the results of the spontaneous QD experiment.

During the positive reinforcement experiment, skinks were rewarded only when they chose the larger option, ensuring that a positive association was established when they actively chose 'correctly'. Overall, the skinks participated in 97% of the total trials, higher than the 83% participation rate during the spontaneous test, and they performed better in the size discrimination trials. These differences in the skinks' performance between the two experimental methods suggest that spontaneous QD may be better suited for testing individuals only once to ensure that the stimuli presented to them are novel and that they cannot learn any rules through repeated exposure. Additionally, because spontaneous discrimination tests only assess spontaneous preferences, they are not as appropriate for studying the absolute cognitive limits of animals. In contrast, spontaneous QD with positive reinforcement, and consequently trained QD, can be used to understand the extent of an animal's numerical discrimination abilities by teaching rules and creating a clear avenue to obtain a reward while minimizing issues with repeated exposure and motivation. Compared to the results of the spontaneous QD test, the skinks had an increased performance following training, indicating that this method can improve a water skink's discrimination ability, allowing researchers to test cognitive limits through learned preferences, as showed in other species such as goldfish and guppies (Bisazza et al., 2014; De Long et al., 2017).

This study is the first to use digital stimuli in a QD experiment involving a nonavian reptile. It shows that digital stimuli could be helpful in further studies of QD, as the skinks responded to the crickets displayed on the iPad screen. The main advantage of using digital stimuli is greater control over continuous variables, allowing the researcher to ensure consistency across trials and guarantee that the same stimuli can be used for all individuals. However, further testing on the importance of different continuous variables may be necessary when designing experiments that use digital stimuli to ensure they are effective for testing the species of interest. Additionally, it is unknown how using a tablet may have affected their visual acuity and whether the skinks struggled to discern individual crickets, specifically when the set size increased during number discrimination.

Future investigations into the visual acuity of nonavian reptile species when faced with digital stimuli compared to the real equivalent will be helpful. For example, the responses of angelfish, *Pterophyllum scalare*, to live and video-recorded shoals found that the fish had weaker responses to videos (Gómez-Laplaza & Gerlai, 2022). This weaker response may also occur in water skinks, resulting in poor accuracy beyond the lowest ratio, but no study has been done on their response to QD with live prey. One possibility is that water skinks may require additional continuous variables, such as total movement, scent or sound, to assist in discriminating prey items, but these were controlled in this experimental design. Research has shown that shoaling fish

perform QD tasks better when they can use total group activity (Agrillo & Dadda, 2007; Gómez-Laplaza & Gerlai, 2012). Although previous research has shown that movement is not crucial for water skinks to hunt prey, our digital stimuli eliminated access to other cues, such as scent or sound (Veron, 1969). As such, because we controlled total movement, scent and sound cues, the water skinks may not have had any continuous information to support the discrimination of more 'complex' ratios.

In summary, we show that eastern water skinks are better at size discrimination than number discrimination, positive reinforcement improved QD performance and they may rely on the ANS for quantity discrimination. If logistically feasible, we recommend complementing spontaneous QD studies with trained QD to maximally understand QD in a species. We also highlight the potential for using digital stimuli to test for QD. The use of digital stimuli has the advantage of controlling for a wide range of variables while presenting consistent stimuli to study animals. Standardized methods will also allow for large-scale comparative studies that have the potential to generate profound insights into cognitive evolution.

Author Contributions

K.N. Rachel Wong: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Simon Jeni:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Benjamin J. Ashton:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Culum Brown:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Martin J. Whiting:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Data Availability

The complete R code and data are publicly available at the Open Science Framework (OSF): https://osf.io/ftkbu/?view_only=894e43e11eaf4aa8b993e2a73f8d15bc.

Declaration of Interest

The authors declare there are no conflicts of interest.

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Appendix

Table A1

P values from the binomial test for side preference for each skink (rounded to two decimal places)

ID	P
MWS002	0.11
MWS003	0.75
MWS009	0.34
MWS019	1.00
MWS022	0.34
MWS024	0.34
MWS026	0.75
MWS027	0.75
MWS031	0.75

Table A2

Number and proportion of trials in which each skink made a choice during the spontaneous quantity discrimination experiment, termed a 'successful trial'

ID	No. of successful trials	Total no. of trials	Proportion of successful trials
MWS002	162	180	0.90
MWS003	125	180	0.69
MWS009	169	180	0.94
MWS019	168	180	0.93
MWS022	153	180	0.85
MWS024	169	180	0.94
MWS026	148	180	0.82
MWS027	126	180	0.70
MWS031	121	180	0.67
Total	1341	1620	0.83



Figure A1. Image of the two crickets used in the side preference test. Both crickets were 2 cm in length.

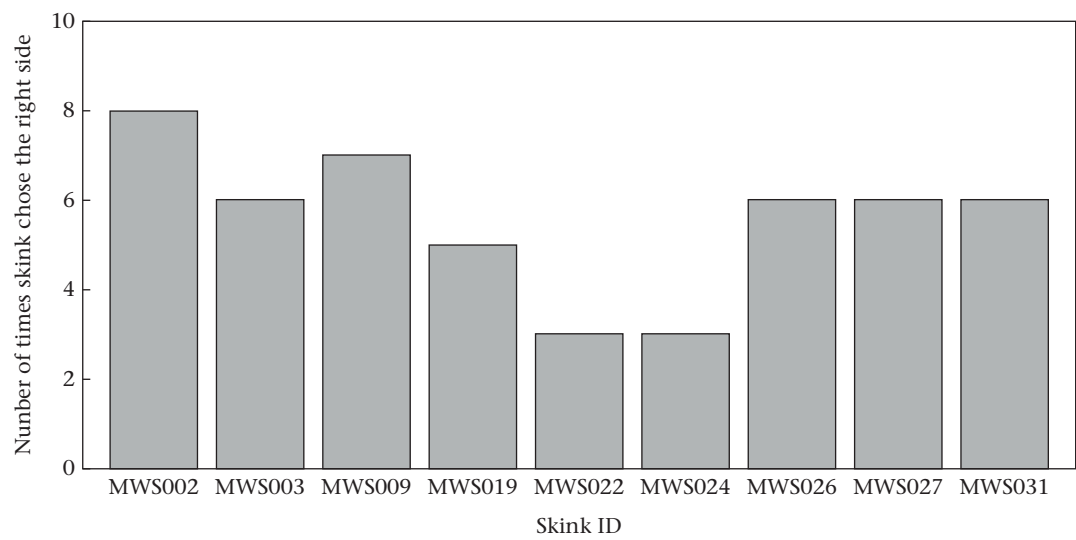


Figure A2. The number of times each skink chose the right side out of 10 trials.

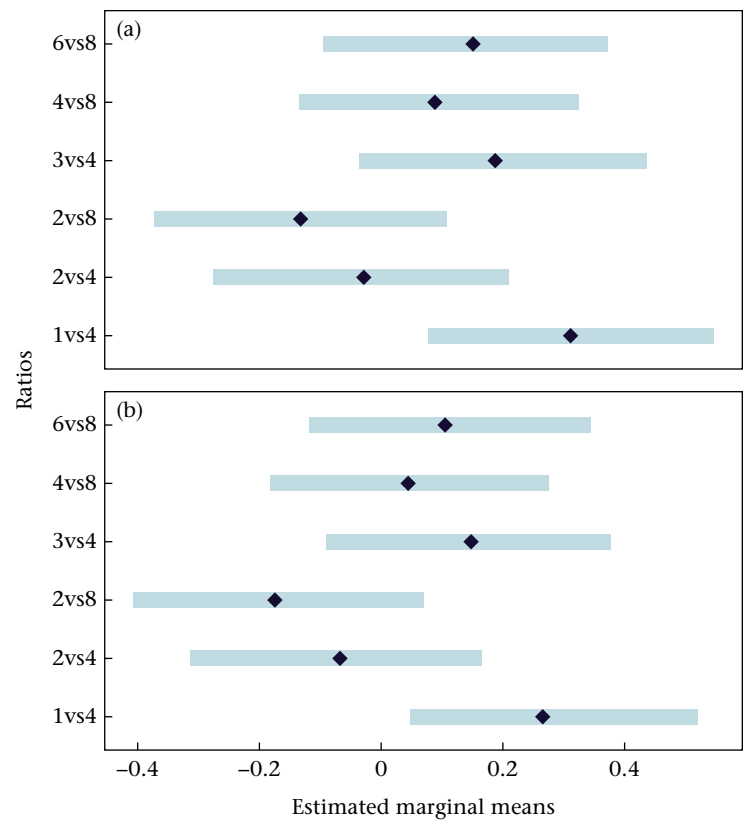


Figure A3. The 89% credible intervals for all ratios. (a) The number of discrimination treatments and (b) the size discrimination treatments for the spontaneous quantity discrimination experiment. Statistical significance was assumed if the 89% credible intervals did not cross 0.